

For rest of today: $E \subset \mathbb{R}^n$ is open, $f: E \rightarrow \mathbb{R}^n$
is loc. Lip., $I_x \subset \mathbb{R}$ is the maximal interval
for any $x \in E$ (w.r.t. $\dot{y} = f(y)$, $y(0) = x$),

$$\Omega = \bigcup_{x \in E} I_x \times \{x\} = \{(t, x) \mid x \in E, t \in I_x\} \subset \mathbb{R} \times E,$$

and $\phi: \Omega \rightarrow E$ is the flow of f .

Lyapunov and LaSalle

(Weak form of)

LaSalle's invariance principle Let $\bar{x} \in E$ be an equilibrium, $U \subset E$ be open,
 $L: U \rightarrow [0, \infty]$ be a weak Lyap func, and $K \subset U$ be a forward
invariant compact neighborhood of $\bar{x}$. Assume that $\{\bar{x}\}$ is the
only nonempty invariant subset of " $L(x)$ "

$$Z := \{x \in K \mid \overbrace{DL(x)f(x)} = 0\}.$$

Then $\bar{x}$ is asymptotically stable, and K is a subset of its basin of attraction $B(\bar{x})$.

pf First note that compactness & fwd inv of K
 $\Rightarrow I_x^+ = [0, \infty) \quad \forall x \in K$.

Next, Lyap direct method $\Rightarrow \bar{x}$ is Lyap stable, so
it suffices to show that $\phi^t(x) \rightarrow \bar{x}$ as $t \rightarrow \infty \quad \forall x \in K$.

Suppose (for contradiction) $\exists x \in K$ s.t. $\phi^t(x) \not\rightarrow \bar{x}$ as $t \rightarrow \infty$.

Then $\exists \varepsilon > 0$ and sequence (t_k) w/ $t_k \rightarrow \infty$ as $k \rightarrow \infty$
s.t. $\phi^{t_k}(x) \in K \setminus B_\varepsilon(\bar{x})$. Since $K \setminus B_\varepsilon(\bar{x})$ is compact,

we can pass to a subseq and assume WLOG that $\exists x_* \in K \setminus B_\varepsilon(\bar{x})$
s.t. $\phi^{t_k}(x) \rightarrow x_*$ as $k \rightarrow \infty$.

We will first show that $\Gamma_{x_*} \subset K$. Fix any $t \in I_{x_*}$.

$\exists k_0 \geq 0$ s.t. $t_{k+k_0} + t \geq 0 \quad \forall k$. Hence

$$\Phi^t(x_*) = \Phi^t\left(\lim_{k \rightarrow \infty} \Phi^{t_k + t_0}(x)\right) = \lim_{k \rightarrow \infty} \underbrace{\Phi^{t+t_k+t_0}(x)}_{\substack{\in K \quad \forall k \\ \text{(by choice of } t_0\text{)}}} \in K.$$

Thus, $\Gamma_{x_*} \subset K$ & hence $I_{x_*} = \mathbb{R}$.

We now show that $\Gamma_{x_*} \subset \mathbb{Z}$. Since $t \mapsto L(\Phi^t(x))$ is nonincreasing, it converges to $L(x_*) = \lim_{k \rightarrow \infty} L(\Phi^{t_k}(x)) = L\left(\underbrace{\lim_{k \rightarrow \infty} \Phi^{t_k}(x)}_{x_*}\right)$.

Thus, for any $t \in I_{x_*} = \mathbb{R}$,

$$\begin{aligned} L(\Phi^t(x_*)) &= L\left(\Phi^t\left(\lim_{k \rightarrow \infty} \Phi^{t_k}(x)\right)\right) \\ \text{(since } L, \Phi \text{ are conti.)} &\longrightarrow = \lim_{k \rightarrow \infty} L(\Phi^{t+t_k}(x)) \\ &= \lim_{s \rightarrow \infty} L(\Phi^s(x)) \\ &= L(x_*). \end{aligned}$$

$\Rightarrow L$ is constant on $\Gamma_{x_*} \subset K$, and hence

$$\Gamma_{x_*} \subset Z. \quad (\text{Indeed, } \forall t \in I_{x_*} = \mathbb{R},$$

$$DL(\phi^t(x_*))f(\phi^t(x_*)) = \frac{d}{dt} L(\phi^t(x_*)) = \frac{d}{dt} L(x_*) = 0.)$$

Since $\Gamma_{x_*} \neq \{\bar{x}\}$ is a nonempty invariant subset of Z , we have a contradiction (since we assumed $\{\bar{x}\}$ was the only such subset).

Thus, $\phi^t(x) \rightarrow \bar{x}$ as $t \rightarrow \infty$ $\forall x \in K$, as desired. $\square$

Cor
(a "global"
LaSalle
variant)

Let $\bar{x} \in E$ be an equilibrium and $L: E \rightarrow [0, \infty)$ be a proper, weak Lyapunov function for $\bar{x}$. Assume that $\{\bar{x}\}$ is the only invariant subset of

$$Z := \{x \in E \mid DL(x)f(x) = 0\}.$$

Then $\bar{x}$ is globally asymptotically stable,

pf Fix any $c > 0$ and define $K_c := L^{-1}([0, c])$. Since L is proper, K_c is a forward invariant compact neighborhood of $\bar{x}$.

Since $\{\bar{x}\}$ is the only nonempty invariant subset of

$$Z_c := \{x \in K_c \mid D L(x) f(x) \leq 0\} \subset Z,$$

LaSalle $\Rightarrow \bar{x}$ is asy stable w/ basin $B(\bar{x}) \supset K_c$.

Since $\mathbb{R}^n = \bigcup_{c>0} K_c$, this shows that $\mathbb{R}^n = B(\bar{x})$. □

Alternate ending:

Now let $x \in E$ be arbitrary. Set $c := L(x)$.

Then $x \in L^{-1}([0, c]) = K_c \subset B(\bar{x})$. Since $x \in E$ was arb,

$$B(\bar{x}) = E.$$
□

↑
(since $\bar{x} \in L^{-1}([0, c])$
 $= L^{-1}((-\infty, c])$ and
 $L^{-1}((-\infty, c]) \subset K_c$ is
open since L is
cont. & $(-\infty, c)$
is open.)

Ex

(damped nonlinear
spring)

Consider $\ddot{x} = -x^3 - c\dot{x}$, $c > 0$. Then w/ $v := \dot{x}$,

$$\left. \begin{aligned} \dot{x} &= v \\ \dot{v} &= -x^3 - cv \end{aligned} \right\} =: f(x, v); \quad f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

E here

$\bar{x} = (0, 0)$ is the unique equilibrium. Is $\bar{x}$ (globally) asymptotically stable?

Define ("energy") $L: \mathbb{R}^2 \rightarrow [0, \infty)$ by

$$L(x, v) = \frac{v^2}{2} + \frac{x^4}{4}.$$

Then $DL(x, v)f(x, v) = x^3(v) + v(-x^3 - cv)$

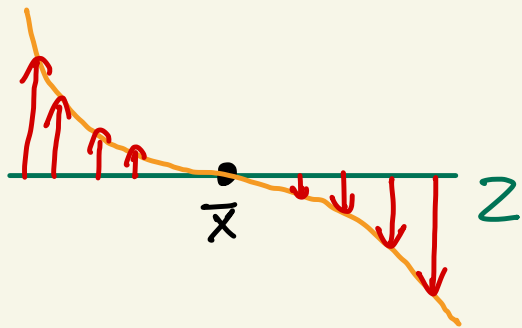
$$\begin{aligned} &= -cv^2 \\ &\leq 0 \end{aligned}$$

$\begin{bmatrix} x^3 & v \end{bmatrix} \begin{bmatrix} v \\ -x^3 - cv \end{bmatrix}$

$\Rightarrow$ L is a weak Lyap func (hence $\bar{x}$ is Lyap stable),
and

$$\begin{aligned} Z &:= \{(x, v) \in \mathbb{R}^2 \mid DL(x, v) + L(x, v) = 0\} \\ &= \{(x, v) \in \mathbb{R}^2 \mid -cv^2 = 0\} \\ &= \mathbb{R} \times \{0\} = \text{"the } x\text{-axis"}, \end{aligned}$$

On Z (= x -axis), $v=0$ and hence $\dot{v} = -x^3$.



$\Rightarrow$ the only nonempty invariant subset
of Z is $\{\bar{x}\}$

$\Rightarrow$ by LaSalle, $\{\bar{x}\}$ is asymptotically stable. ✓

What about GAS? If we can show that L
is proper, then $\bar{x}$ is GAS (by the Thm of lec 24).

Note that $\partial \mathbb{R}^2$ is empty (since $\partial \mathbb{R}^2 := \text{cl}(\mathbb{R}^2) \setminus \text{int}(\mathbb{R}^2) = \mathbb{R}^2 \setminus \mathbb{R}^2$).

Let (x_n, v_n) be any seq s.t. $\|(x_n, v_n)\| \rightarrow \infty$ as $n \rightarrow \infty$.

Then
$$L(x_n, v_n) = \frac{v_n^2}{2} + \frac{x_n^4}{4} \geq \frac{1}{2}(v_n^2 + x_n^2 - \frac{1}{16}) = \frac{1}{2}\|(x_n, v_n)\|^2 - \frac{1}{32} \rightarrow \infty$$

Since $x^4 \geq x^2 - \frac{1}{4}$,
So L is proper by above Note and the Rmk of Lec 23.

So the Corollary of LaSalle above $\Rightarrow \bar{x} = (0, 0)$

is globally asymptotically stable!

(More generally):

If $k, l > 0$ and $2k > l$, then $\exists C \in \mathbb{R}$ s.t.
 $\inf_{x \in \mathbb{R}} (x^{2k} - x^l) \geq C$,
so $x^{2k} \geq x^l + C$
 $\forall x \in \mathbb{R}$.

